INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

APPLICATION SPRAY EQUIPMENT FOR NONSKID COATINGS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the general requirements for the design features and salient qualities of spray equipment capable of applying mixed nonskid deck coatings designed for naval applications by spray transfer technology. The spray equipment provides a means of combining the resin, hardener, and aggregate in a single container and applying the combined product through an atomizing spray gun.

2. SALIENT CHARACTERISTICS.

2.1 Equipment. The spray equipment covered by this CID shall have an open hopper that feeds a progressive cavity pump for fluid transfer to an atomizing spray gun that shall be procured as a single unit. As a minimum, the spray equipment shall include the subassemblies and components listed below and discussed in the following paragraphs.

a. Design and construction
b. Piping
c. Dispensing pump
d. Electronic pump motor control box
e. Diagnostic motor monitoring system
f. Electrical junction box
g. Pneumatic dispensing on/off flow control
h. Electronic air flow monitoring system
i. Spray gun
j. Diffuser spray head
k. Mobile support frame or platform
l. Operating instructions and repair manuals

2.2 Design and construction. The spray equipment shall be a complete and fully functional assembly consisting of a pump and associated control systems mounted on a mobile support frame or platform. The assembly shall include a minimum 25-foot outlet material hose, a minimum 25-foot atomizing air hose, and suitable connections to transmit nonskid from the hopper to the spray gun. The assembly shall include a spray gun and a minimum of two spray diffuser heads. The pump and spray gun shall be designed for rapid assembly and disassembly through the use of press fittings, quick disconnects, cam locks, large threaded fittings, and inexpensive disposable parts. All non-disposable parts, finishes, and hardware shall be suitable for exterior exposure in a marine environment and shall not be degraded by exposure to epoxy and polysiloxane nonskid cleaning solvents.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Design pressure. All piping, hoses, fittings, valves, and other pressure containing components shall be rated for service at or above the maximum design pressure of the particular system in which they are used and as defined by the spray equipment manufacturer.

2.4 Dispensing pump. The dispensing pump assembly shall consist of a quick disassembly hopper and pump system to contain mixed nonskid; a progressive cavity pump; and an electric motor, gearbox, and connecting rod assembly to power the pump. All components shall be mounted on the support frame or platform using threaded fasteners to allow efficient dispensing pump system disassembly and repair. The dispensing pump shall have an upper and lower hopper system that can be separated without the use of hand tools, with a minimum capacity of 6 gallons that is adequate to contain a mixed 5-gallon nonskid kit. The stator pump assembly shall be easily detached from the lower hopper exposing the seal and stator pump for easy cleaning/replacement and constructed of an inherently solvent resistant material. The hopper shall have a minimum 23-inch diameter opening to allow easy addition of material from 5-gallon cans into the unit during continuous nonskid spray application. The dispensing pump shall be of progressive cavity design and consist of a helical metal rotor and rubber stator that pumps material through the fluid hoses to the spray gun. The dispensing pump materials shall be wear resistant to allow the pump to dispense a minimum of 1,000 gallons of material without requiring any maintenance beyond normal solvent cleaning. The rotor shall be connected to the gearbox shaft by a mechanical seal and connecting rod. The mechanical seal shall be wear resistant and designed to allow 1,000 gallons of material to be pumped without requiring any maintenance beyond normal solvent cleaning and lubrication. The pump drive motor shall be a 110- to 120-volt (V)/15-amp (A) electric motor with a minimum of a 1 horsepower rating. The pump drive motor assembly shall include a continuously variable, electronic motor speed control that allows manual pump rotor speed changes from 1 to 200 revolutions per minute.

2.5 Electronic pump motor control box. The electronic pump motor control box shall allow the operator to switch motor power on/off, set selectable pump motor speeds to vary the delivery volume, and shall include appropriate sensors and microprocessors to provide an automated diagnostic motor monitoring system.

2.6 Diagnostic motor monitoring system. The diagnostic motor monitoring system shall continually self-test the electric motor condition and shall provide visual and audible warnings and indications of the following:

a. Run mode
b. Stop mode
c. Stand-by mode
d. Short circuit
e. Ground fault
f. Motor overload
g. Under voltage
h. Recovered under voltage
i. Recovered over voltage

2.7 Electrical junction box. The junction box shall be powered by 110 to 120V alternating current (AC) and require no more than 15A for normal operations. A 30- to 50-foot long electrical power cord attached to the electrical junction box shall provide system operational electrical power. The electrical power cord shall terminate in a standard, male three-prong grounded plug and shall be fabricated from AWG-12 three conductor wire. The electrical junction box shall control system power and shall include on/off switching with associated indicator lights and electrical system fusing, and shall control pneumatic switching of pump motor on/off operation via signals from the pneumatic fluid dispensing control system. It shall also include a coating-dispensing pump motor jog switch and power transformer for the electronic air flow monitoring system. All wire, indicator lights, and displays used shall be inherently suitable for exterior service and shall not be degraded by exposure to nonskid cleaning solvents. All components shall be contained within a vapor tight and watertight NEMA 4 rated enclosure.

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2.8 Pneumatic dispensing on/off flow control. The pneumatic dispensing on/off flow control system shall enable start/stop operation of the dispensing pump as well as start/stop operation of the material flow by means of a pneumatic valve. The pneumatic valve shall be operated by a guarded push button air valve that shall provide enough closing force to stop the nonskid flow. The pneumatic valve shall be configured such that when the push button valve is pressed down, the pump motor starts and the coating dispense valve opens to allow material flow through the hose assembly to the spray gun.

2.9 Electronic air flow monitoring system. The incoming air supply that provides atomizing air to the spray gun shall be routed through an electronic air flow meter that displays the air flow rate in cubic feet per minute on the electrical junction box. The incoming air supply shall be routed through an in-line filter and coalescer to ensure that air delivered to the spray system does not contain moisture or oil as determined by ASTM D4285, Standard Test Method for Indicating Oil or Water in Compressed Air. The volume of air flow to the spray gun shall be controlled by an inline needle valve located on the mobile support frame before the flow meter.

2.10 Spray gun. The spray gun shall be fabricated from lightweight materials that are inherently suitable for exterior service and resistant to nonskid solvents. The spray gun shall include a spray diffuser head, pneumatic flow valve, atomizing air on/off valve, and the pneumatic fluid control on/off push button. The on/off push button shall open and close the fluid control valve and start or stop the electric pump motor. All pneumatic control tubes, the spray diffuser head, material dispensing hoses, and atomizing air hose are to be connected to the spray gun using quick-connect fittings. When the fittings and hoses are disconnected, the spray gun assembly shall weigh no more than 10 pounds.

2.11 Diffuser spray head. The diffuser spray head shall be capable of spraying mixed nonskid coatings for a minimum period of 30 minutes without clogging, while producing a conical spray pattern for uniform coverage. The diffuser spray head shall include design features to allow rapid clearing of clogs. The diffuser spray head shall support a minimum material flow rate of 0.5 gallon per minute during continuous spray operation. The diffuser spray head shall have a minimum service life of 300 gallons of material without repair or component replacement. The diffuser shall include design features that dispense a uniform, conical pattern of material without evidence of fingering, clumping, or spattering. Air flow to the diffuser spray head shall be controlled by the airflow needle valve. The diffuser spray head in conjunction with the electronic air flow system shall enable the nonskid to be applied at a minimum rate of 10 square feet per minute. The appearance of the sprayed nonskid shall be uniform with pronounced aggregate peaks for roughness and with complete coverage of the primer as determined by visual inspection without magnification.

2.12 Mobile support frame or platform. The spray equipment components including the pump and motor assembly, electronic motor control box, air flow regulator, and junction box assembly shall be securely mounted on a mobile frame or platform that is designed and finished to resist corrosion in the marine atmosphere, to be inherently suitable for exterior operation, and that shall not be degraded by exposure to nonskid cleaning solvents. The frame shall include a minimum of four swivel casters with a minimum roller wheel diameter of 4 inches to allow manual maneuvering of the system on ship decks. A frame rail shall surround the platform and components to provide hand holds for equipment positioning and to prevent damage if the unit is overturned.

2.13 Operating instructions and repair manuals. The spray equipment shall be delivered with operating instructions and repair manuals that describe basic equipment operations, safety features, and repair processes. Manuals shall include descriptions of processes for ordering and installing repair parts. Hard copies of the manuals shall be supplied with each unit and additional electronic copies shall be available in accordance with the contract or purchase order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

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4.2 Market acceptability. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID:

a. The manufacturer producing the item shall have a fully functional prototype available for demonstration that meets the requirements of this CID.

b. The company producing the item shall have been producing and maintaining commercial paint spray equipment for at least 10 years.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

6.1.2 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

6.1.3 NAVSEA Publications. NAVSEA publications may be obtained via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number.

b. Product conformance provisions.

c. Packaging requirements.

d. Quantity required.

6.3 Key words.

Dispensing pump

Hopper

Nonslip

Progressive cavity

Spray gun

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MILITARY INTERESTS:

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Army – AV
Navy – SH

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Army – MI, MT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
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